Skin Aging

Extracellular Matrix Changes

With aging, the level of **fibulin-5**, an extracellular matrix protein that functions as a scaffold for elastic fibers, appears to decrease before other changes are observed, suggesting that loss of fibulin-5 is a marker for skin aging.

The ground substance components—**mucopolysaccharides**, **glycosaminoglycans (GAGs)**, and **proteoglycans**—are decreased relative to dry weight or collagen content of the skin, especially **hyaluronic acid**, possibly due to decreased hyaluronan secretion or reduced extractability. Aging also alters GAG composition and their binding to elastin, impairing molecular drainage into lymphatic vessels. These changes may adversely affect skin turgor, as proteoglycans can bind up to 1,000 times their weight in water, and also influence collagen fiber deposition, orientation, and size.

Mechanical Properties of Aging Skin

Age-related changes in the mechanical properties of adult skin include: - Progressive loss of elastic recovery, consistent with gradual degradation of the dermal elastic network - Marked prolongation of the time required for excised skin to return to its original thickness

In vivo ultrasound studies reveal age-associated differences in water distribution within the dermis, impacting dermal pliability, resilience, and elasticity. Overall, the aging dermis becomes increasingly rigid, inelastic, and unresponsive, with diminished capacity to adapt to injury or stress.

Subcutaneous Tissue, Muscles, and Bone

Facial Muscles and Fat Redistribution

Like other striated muscles, facial muscles accumulate **lipofuscin**, the "age pigment" indicative of cellular damage. Combined with diminished neuromuscular control, this contributes to wrinkle formation.

Subcutaneous fat is depleted from specific facial regions, including: - Forehead - Preorbital area - Buccal region - Temporal region - Perioral region

Conversely, fatty tissue increases in: - Submental region - Jowls - Nasolabial folds - Lateral malar areas

Unlike the diffuse fat distribution in youth, fat in the aged face redistributes under the influence of gravity, leading to sagging and drooping of the skin.

Facial Bone Remodeling

Facial bones undergo age-related reduction in mass due to resorption, particularly affecting the: - Mandible - Maxilla - Frontal bones

This bone loss exacerbates facial skin droopiness and diminishes the clear jawline contour seen in young adults, blurring the demarcation between the jaw and neck.

Hair

Graying of Hair

By the end of the fifth decade, approximately half the population has at least 50% gray (white) scalp hair, and nearly everyone experiences some degree of graying. This results from progressive and eventual complete loss of **melanocytes** from the hair bulb.

Melanocyte loss occurs more rapidly in hair than in skin because: - Hair bulb melanocytes are highly active during the **anagen phase**, producing melanin at maximal rates - Epidermal melanocytes remain comparatively inactive throughout life

Hair graying specifically reflects depletion of **melanocyte stem cells** in the hair follicle bulge, potentially due to disrupted interaction between two transcription factors: **microphthalmia-associated transcription factor (Mitf)** and **Pax3**.

Scalp hair tends to gray faster than other body hair due to its higher anagen-to-telogen ratio.

Structural and Functional Hair Changes with Age

Aging is associated with: - Modest decrease in the number of hair follicles, due to atrophy and fibrosis - Increased proportion of follicles in the **telogen** (resting) phase - Hairs that are smaller in diameter and grow more slowly

One hypothesis suggests that loss of melanocytes and impaired melanosomal transfer increase oxidative stress in metabolically active hair follicle keratinocytes, compromising their function and survival.

Hair Loss and Balding

Balding primarily results from androgen-dependent transformation of dark, thick terminal scalp hairs into lightly pigmented, short, fine **vellus-like hairs**, resembling those on the ventral forearm. This process affects men more frequently and severely than women.

In post-menopausal women, hair loss is also linked to: - Decreased estrogen levels - Altered estrogen-androgen ratio